

ActInf Textbook Group ~ Cohort 7 ~ Session 17 (Chapter 6) ~ 2/27/2025

TextbookGroup/ParrPezzuloFriston2022/Cohort_7 | Cohort 7 ~ Session 17 (Chapter 6) ~ 2/27/2025 | 1:02:18

all right welcome thank you everyone for
joining it
is the first early session for cohort 7
in this second phase of
activity so anybody who uh wants to sort
of just check in
with uh that I guess we could have done
before starting recording but just want
to check in where are we at in this
cohort as we head into the second half
of this textbook in people's view or
what would they like to do for this
phase of activity for the kind of second
half for this cohort and then we will
turn to chapter
six um maybe
ali uh and then anyone else who wants to
raise your hand or just go for
it uh okay so uh chapter six actually
marks the um uh beginning of the second
part of the
textbook uh as we probably already uh
know the book organized into two
different uh parts so part one is mainly
theoretical and part one part two uh
basically deals with more practical
applications of active
inference uh
so as as we saw in the first part of the
uh textbook um the theoretical
Foundation of active inference is
actually work in progress so uh it's not
something uh that that has been
uh uh completed uh in order to applied
uh as a kind of uh comprehensive uh

modeling tool but at the same time uh
the second part of the uh textbook
provides very promising road maps uh
to um towards how we can apply even
these uh
tentative foundational work um in order
to build uh some real
World models of the intelligent agents
or cognitive agents uh so I think that's
important to keep that in mind because
uh in many textbook groups uh I've seen
people uh somehow uh raise this issue as
to well it's not as comprehensive uh as
we thought as as a modeling tool so uh
since the public a of this book U
obviously active influence research has
progressed a lot uh but
even the uh the theoretical foundations
provided in the uh first part of the
book uh I believe provides enough
material to um at least build some
uh Toy
models uh that can capture some of the
real world element
of a real um a real world agents or at
least cognitive
agents thank you Ali great anyone else
want
to share with their ad as in this uh as
we head into this part of the
textbook yes so uh I think I would like
to do that um I missed the first half of
this um group uh so I'm I'm asking a
question which belongs to chapter two I
think or chapter three I don't know um
and so my apologies for that and I don't
want to
hijack the discussion here uh and I've

put my question also in the in the chat
but basically comes down
to um having tood the role of PR in um
active inference as having a dual role
which is one kind of the intuitive role
for me at least being vely familiar with
bence inference is you know encoding
prior beliefs and also um having a kind
of indicator role of
encoding uh preferences and I think this
comes out of uh discussion of
um ocular motor Behavior as well as
other things and I was just wondering if
you have discussed this at all and uh if
I've yeah if a brief comment on this
would be great but if you say well you
should have you know attended the
previous sessions and I'm fully happy
with that as well and then we'll just
tag along for the right for
now yes um there's
a slide in this the point raised in this
paper um that points
to Prior being used as a um preference
to drive pragmatic behavior in lie of
for example what a reward function does
for reinforcement learning because when
things are framed in terms of likelihood
directly
rather than projecting onto a explicit
rewarder
utility uh function only just working
with a base joint
distribution
um then the prior can be seen as
preference
constraint so it's like the preference
for blood sugar has a different

connotation and um function and
everything than one's felt psychological
preference for something or revealed or
stated

preference but

um at

least um from a modeling perspective

it's more similar than

not okay so I'll I'll have a look at uh

those reses thanks

so okay

Jerry yeah I think as as we're going

into at least my area of interest and my

questions are you know based in my field

of work which is marketing and

Communications and just how to apply

this to you know um consumer Behavior

you know to to actually build some

models that use these principles and

these

Frameworks um so that we can have

replicable studies um within this type

of a

field great well chapter six is a great

place for that because uh uh it is a

recipe set of

questions with more in common than uh uh

with just

general uh statistical experiment

planning and systems engineering and

systems modeling

broadly uh with a connection in specific

ways to get to the kind of generative

models that are described in basically

chapter 4

and there's a lot of uh distance

between where the this uh current text

from 2022 will get you and the code

running in your
hands uh but that's a ground that is
also rapidly being more traversible
too and uh and so building out examples
so that they're specific
uh just you know there's there's a big
gap there but that is it represents um
in a sense just how many degrees of
freedom there really are in
probabilistic
modeling yep that makes sense and so
yeah I'm excited to dig into it it
definitely seems like it's an area that
uh is ready to be
explored cool so I wanted to share this
uh one little experiment this is from
earlier today it's linked in the chapter
page so I took the text of chapter 6
just plain text put it into this PL tax
repo used cursor
IDE to restructure it and obsidian link
it so now here it is um with probably
some light uh you know
edits
but has these links out to all these
other
topics also this
repo you can just put a new file and say
write a PDP type model for this because
it has this whole knowledge scaffold
that that we can also all contribute to
so I think as the uh
years and the application Niche
changes chap uh the textbook and chapter
six like Ali said it will always be
interesting to consider how it's like oh
yeah just these matrices alone are kind
of like a good representation of

modeling this or

that

uh all the way on through however that
gets friction and a lot of degrees of
freedom in application when there are a
few standardized open-source methods and
it's unclear how to necessarily

Implement just an equation into an
actual program they like um or there can
be multiple it just like oh why is there
not just one KL Divergence function or
like why don't we just have a
variational free energy function there's
sometime it sometimes it gets very uh
data form

specific computational environment

specific mat lab Julia all these like

things that come up in the

application so I think that's one of the

biggest changes between

2022 um as uh some will remember in
terms of like we'll think okay before RX
and

fur pmdp mat

lab now better augmented coding and

knowledge engineering tools code

generation

tools

um

helping with some of that especially if
we uh can collaborate I believe and have
some people working with these code and
program synthesis approaches to the
structure learning and kind of uh
sketching out more the Tex tree of the
spaces of generating
different aspects of specific
uh generative modeling uh

passages and then also people who are just like wanting to model or do inquiry into a specific like yeah do there have more squirrel crossing the bike path on Tuesday those kinds of uh test experiments so um but this is the application part it's almost like the textbook coming back to domain uh even though that kind of happens in chapter 5 to an extent so I hope that in this phase for we can uh possibly return to some of the code questions that have come up over the years like having code examples for um you know chapter 7 in chapter uh some of which we already have more uh Advanced things open source so it' be a subsetting not like a making but also these things just from being specified can be made very quickly to a first pass so if we want to to um develop the code and also the questions around going through application that would be really useful maybe maybe I could share my um project that I'm working on and then when it comes up relevant uh I'm Andre from Lithuania Andre kowas does anyone want to start with a specific question about chapter 6 or the book first and then then let's do projects but not um begin with it thank you address what should we really do in synchronous asynchronous for the textbook the 2022 textbook is a great art artifact

understand and learn from I mean any
section or any question can be
useful um though really I am
I

just nice nice

drawing hey Daniel um yeah at least
you're just asking questions there like
the the one up there about affordances
is always of interest to me as well as
you know maybe how like variational
script might come into play if that's
applicable as

well yeah what what about

it um I

mean I think at least from from my
perspective um there was some stuff I'd
read Around creating fields of
affordances which I felt kind of tied
from a consumerism perspective
to how Brands go out to market for
instance um and so maybe the question is
more around like not just affordance
generally but that idea of like how do
organisms in this model or how would we
test

um how agents select between the field
of importances through those either I
don't know if that's through policy
selection there that's my assumption but
um you know still newer to thinking
through this from a like actual
application

perspective does anyone want to say
something to

okay the first question in the for four
question recipe here is what's this
system are we modeling and then if we
were thinking about an

individual uh Mouse in the tmas like
what's going to come up in chapter 7 in
discrete time it's like okay are we're
going to model it with four locations
starting left right and
down uh or are we going to model its
footsteps so if you model footsteps then
there's going to be the the explicit
you're going to have some data structure
that represents the footsteps location
like data about it Andor
simulated distributions about it but if
you're going to just say well not it's
not a footstep modeling project it's
just a a location for state that which
is what's going to happen in chapter
7 so um just from this sort of Applied
angle which system are Remodeling and
that's kind of reflect that the earlier
comment about this having more in common
especially early in this process with
systems modeling and systems
engineering it's just like okay the
organism is a very complex Locus person
at a game machine or something but but
um like the complexity of how hormones
interact with uh you know sound and all
this or something or microbiome those
are all part of the real environment SL
situation SL system but you might be
able to get away with a 3X3
Matrix for just three uh states of
vigilance low medium
high because those kinds of modeling
uh that's just oh
all models uh map
territory all those kinds of topics that
that come back and but it's like right

it's a it's a model of this it's an
inferred parameter it's a risk factor
for x or it's just a mutual information
estimate on percentile on this in our
model so it that it that's why this
whole process where chapter 6 actually
gets one isn't necessarily some um G
paradigm shift on their modeling view
per
se that you couldn't necessarily also
have from doing a regular gen
generalized linear
modeling which can also be very
sophisticated and deal with higher
order um variance and causal relations
it's kind of
like generalized probabilistic
modeling combined with cognitive syst
systems approaches to systems
engineering okay yeah that makes sense
in the footstep model when you brought
that up I think that kind of resonates
for me because we'll talk a lot about
customer Journeys and one thing I've
been thinking about is just you know
using the T the term consumerism like I
don't know that we think about it the
right way in my industry and so I've
been looking to biology a little bit for
a better analogy and so like predatory
behavior for instance like the process
of identifying and detecting and
consuming and then processing something
I think is an interesting corollary to
what we probably ought to be doing when
we talk about you know communication and
purchase and stuff like that so that's
that's interesting and helpful um and

maybe that's something I can dig a little bit more into is that idea of more modeling footsteps versus locations which I think we tend to try to model the end the location when we ought to be modeling like the steps towards that yeah so the the hierarchical models at least in principle which is kind of where the second half of the Tex goes is is um where you have slower inference models you know and then a faster ones and then uh it's just a project or situation

specific empirical question about how much compute and what kinds of specific models you're going to make um so so this let's continue on as people if if they want to bring in their um you know how they're seeing that or some other section of the of six or we can go to the questions but just yeah this first question is like is like okay we're

modeling everything about the Biometrics of the person or just this those are such different questions but those are questions that are also just dealt with everyday in animal behavior cognitive modeling okay but here's where we start getting more into some of the technical analytical

details of the structure of these models that relate to making some more committed model

decisions not not just agreeing that okay we're we're just interested in um this question about this system of interest um so that that the next three

questions are the three practical challenges these can be specified um like they go into more detail in chapter 6 and there's there's more sort of links into guides and experiences people can share and and just try and explore uh through but basically it's like keeping in mind that the textbook here uses generative model to describe the agent type uh unit and the generative process to be the environment um whereas also generative model can be used just to be the total um model specification basically the three um areas of questions are how do you want to deal with time um model it discretely like or in continuous time uh these are questions common to other time series modeling approaches um even if you choose continuous time you know how will time discretization on if Will that matter for your data or for your compute um what is going to be the structure um in hierarchy

[Music]

um laterality like just message passing is it just like are you modeling one abstract decision maker making one sequential response to one put data just as an intuition pump or is it doing like this data to here and wanting this output so that's just sort of the um getting to specifics about the model

structure which can again just be either specified

uh for from Simplicity like just be often the models are just a single uh PDP partially observable Markov decision process and that that um can't necessarily be directly picked up to work on a large data set or something like that but for certain purposes it might be all the research question needs at that stage but um again that's kind of a separating like okay we know the systems have hierarchical depth so what parts of multiscale complexity are modeled so that's just a common modeling question and then what is the time how do these different agent or generative model specifications themselves deal with time so just some it's uh things that come up but these are more in common with uh at this point like statistical modeling causal inference machine learning AI also these kinds of questions would come up here is where it starts to get more into um some of the uh basian

uh degrees of freedom of modeling like are you going to model a parameter as just fixed like a spike in the probability space like not learnable you know how learnable how variable and how what are you going to do with how you specify and update and

fix how learnable and how variable are
you just going to say from this previous
study we know the variance of this trait
to be 10 so we just fixed the variance
at 10 or we learned it but um you know
if everything is learnable there's just
the Looms a greater problem just like in
uh uh reward learning when when they get
into meta learning and all this how
learning how fast to learn to learn and
all the sort of things
is um I mean the larger the state space
learnability of the
the of the kind of Target Model you make
an optimization challenge by making it
so open-ended by dealing with large data
by dealing with complex
data maybe there are some
structures of probabilistic models that
have excellent computational performance
on certain
situations but that's a hope and a
preference and uh is not the case that
that that uh you know any
given uh probabilistic model would be
performant in any given way on any given
data
set however there might be some that are
really really
effective or or the huge families that
are adequate for certain
uh uh processes but but but we're we're
getting into statistical
modeling so that's kind of
from coming coming to this from an
empirical biology behavioral modeling
perspective and seeing how mostly here
we're just talking about

how some this could be said and probably
many parts are said across
different cognitive and
behavioral modeling works
but it's working within a
specific probabilistic modeling
framework that has connections to the
free energy
principle um sort of path of least
action
approach coming from The High Road
chapter six sort of brings us back to
the low road and building up the
specific
systems that um also have the extension
into the
implementation Methods at least uh some
as we sort of explored a little earlier
but it
um getting
from the
uh questions about the specific
system to the probability distributions
in families and mod model that that meet
up in the Middle with the sort of high
road
approach is basically where chapter 6 is
but there's no it's like these questions
are all common to statistical
modeling efforts and there's other
things I'm sure people here too could
add of just the sort of um her istics
and pitfalls and important questions for
interpretation of complex systems
modeling another sort of I mean
another
uh practical question is
like um where what is the what there's

different ways to do projects and and
research and everything but does someone
want to just imagine a model where
there's a
10 time Horizon and a five of this and
it takes in this data it's like is it
just being
sketched as part of a
thinking process
or is this
wanting to be used for doing statistical
modeling because the sometimes there's
moves that are
possible in just sort of drawing
connections on paper that might be very
uh there's that might be like hard
within current
implementation
approaches so in the examples of of the
code we kind of are building out
um motifs that are
useful but the the way that people are
thinking about
systems um is often very
nuanced and so it's I I I mean I I hope
people can in the
second uh phase here like let let uh
figure out how to make it useful also
just discuss what do we really want to
do with these meetings with recorded and
unrecorded part
because we we we can figure out
um other ways to have our shared
attention and work I don't
want to uh feel
like returning to this and just
re-recording going back to the same if
that's not the most useful format

this

to kind of break the to kind of break
the silence here I'm not PR sure because
like what um what what do you usually do
like what is your usual um mode of like
navigating uh the second half of the
book like the Practical part of the book
um like for instance someone uh like I
think Andreas was about

to sorry I was about to talk about his
project I think hearing about uh like
the projects that people are working on
like how they're implement all this
stuff that would definitely

be sorry that would definitely be
interesting for me um also for my own
project like I don't have anything
specific yet but for the coming meetings
I think it would definitely be
interesting to um to discuss the
specifics of that but I'm I'm not sure
what other things you you would yourself
suggest J Daniel or like what has worked
for uh previous previous meetings but
that would be interesting for

me thanks you yes okay we can return
that's a good suggestion for Andreas um
in terms of what has been useful people
can look through the previous chapter
6 uh

meetings but I I think what would be
very useful is

to

uh connect specific questions and ideas
that people have to segments of the text
if only as an exercise in anchoring and
and Scaffolding off of the
2022 text um maybe just as a jumping off

point in moving to other

resources

then

uh working through these questions and

figuring out what other questions they

asked along the way

to get to they wanted to go with a

probabilistic model that that they

um could share uh back and uh connect to

implementations uh tables that that we

have to the textbook

examples so sort

of it to if we want to make models in

given domains let's make them over these

coming months according to this recipe

Andor making our own but uh at least

taking this

recipe uh seriously as or even as a

subset of things that we do

Andreas how about just go for your

question let's see where it goes yeah uh

so um um to introduce myself I'm Andrew

kowas from Lithuania I learned about

active inference a year ago from Daniel

and I've had many conversations with him

I've read the first six chapters and so

I'm attending at this time U every week

uh and so what I'm interested in

modeling um I'll say generally and then

specifically generally I have this uh

I've been documenting my own human

expeance uh as a conceptual language and

so

I'm uh one way that comes out is like I

I feel like I live my life through three

Minds there's an answering mind a

questioning mind an investigating mind

so like if I raise my hand there's me

that knows what to do and I just tell myself to raise it and I don't have to think very much about how it's being raised or where it's being raised so and similarly like there's all kinds of answers that come into my head so that's the unconscious um these answers and then there I have another part of mind that's this chattering language uh words or or concepts but I think of them as questions so like when I think of like the word horse it's really a placeholder for what the answering mind might tell me as a horse and so I have this conceptual language and then I have um this full-fledged Consciousness which I call the investigating mind willful cognizant

deliberate which basically tries to um line up the other two because get the same information in two different forms so the unconscious knows all these things the unconscious doesn't know them and so can have the information that two and then like with conom okay

I apologize can you hear me yeah continue I apologize on so I was talking about this investigatory mind which aligns the other two what we know what we don't know and then like with Daniel conman's Thinking Fast thinking slow it's able to decide hey here I should just think fast and and where here is here I should think rationally you know so go with my Intuition or should go rationally I'm trying to model that so specifically how could I model

especially and I think there's a
language for each of those like so
there's a language of how things come to
matter for the unconscious there's a
language of how meaning arises like
verbalization for the conscious and
there's a language for
how
okay not sure if his connection
staple um
yeah Danel to the last part he was
talking about I I think I have a
question maybe tack on to that about
like when he's talking about system one
system two yeah does that in some way
tie to like how we would try to model
novelty versus habit um in terms of like
you that's something I've been thinking
about as well like a lot of behavior
that we notice in my field is is habit
but we're trying to catch attention I
think through novelty
um to train something new in there yeah
like um
to that um and also a little bit about
between habit and deliberation which
Andre I hope we'll start to get at some
of these points that that we can
continue to um work on and develop
in here in chapter 5 so not to go back
but um chapter 5 reviews the some of the
key model motifs from the Maman
neuroanatomy cognitive modeling
literature um studying people's brains H
that being where friston and others have
focused a lot of their work um so as
sort of like well this is sort of what
it looks like to study a complex system

uh like the brain on this so here are two different um extrema modes of how policy priors like how apt one is to do something before receiving the next observation and uh policy inference uh getting to a policy posterior which action is sampled from so it's like the differencing of what was apt to happen before to what was apt to happen after for action and these are the distributions that are calculated during policy uh inference planning as inference actually implementing planning agents you will have one um or the other or a modulated blend of these as as it turns out to be for the brain for policy inference on the left is a habit passed through mode which is like cached action preferences which can be conditionally set on the right is the expected free energy calculation which is uh discussed more in the first half of the book and we can look at implementations in PDP or in uh other implementations in this phase of activity but this is where specific counterfactual policies of a given temporal Horizon are rolled out like policy research and evaluated in terms of expected free energy or free energy the expected future um so explicit policy roll out reweighted energy based learning consideration of epistemic plus pragmatic value of policies um know large computational

blowup just like any other tree based uh
policy search so that's where um
secondary
teristics like um bounded TR searches
and policy switching and things like
that uh come into play for for making
these models large but often again it's
only a simple one with four policies
being considered or something or 16
policies being
considered but but this has that sort of
multi
combinatoric um computational complexity
if explicit probabilistic or to the
extent explicit probabilistic rollouts
are
used um and then on the left is like
basically just a pass
through so broadly that connects to the
thinking fast and slow and I think I
think some papers
have made some um points but it just
sort of
a it's a functional uh stratification in
cognitive
systems that and then that is being here
modeled by one switching parameter not
map and the territory all that it's like
could there be two switching parameters
but that's the
whole experimental
biology question
and I would jump in to say like with
marketing um there's very much the
opportunity like you can sell to
somebody's reflexes or you can sell to
somebody's rationality and if you're a
buyer you know you kind of want to have

both satisfied you want to feel good um
instinctively and you want to feel good
in terms of you know your rational
judgment uh and so I apologize for my
bad connection but I wanted to say um
what I want to model specifically I
think would be um uh this language of
how things come to have meaning which is
relevant for the rational mind so
because the mind that just works
instinctively it just doesn't need to
know all these things but once you
introduce this second rational mind it
needs to think both you know has to kind
of be aware of both ways and kind of do
do its best with this other
mind so I thought well mathematics is
something where things don't really
matter but the meaning arises all the
time and I thought I would look in
particular to a very narrow field which
would be triangle geometry and it turns
out in that field there's an
encyclopedia of triangle centers and
basically like everything interesting
about triangles is basically like a
center of a circle so you can build that
center of a triangle so to speak in
60,000 different ways but to to focus on
the very simplest way um if you put a
circle in a triangle it doesn't matter
what the shape of the triangle is but um
if you expand it to the maximum size
there will be three points that it
intersects that it touches on the
triangle and the property will be
because the radius of the circle is
fixed that from the center of the circle

it will be equally positioned you know distance-wise to each of those three touching points so that's called the incircle so one way is to try to imagine these two minds as communicating that problem so the unconscious mind kind of imagine it it places the circle in the Triangle you know you're giving a random triangle and you're kind of like blind so the unconscious mind places a circle in the triangle and it's allowed to expand the circle until it touches something and it has to stop and mathematically it'll only touch one side basically like probabilistically it's impossible to touch more than one side at a time and then the conscious mind will say okay I want you to move but let's say it can't it can give an instruction to to move but uh the motion will be random in some random Direction so or I guess it has to move it has to move away from the it has it only has one direction it has if it doesn't want it has to move away from the line so it just moves away from the line at a certain speed let's say that's based on the radius proportional to the radius and then it will the conscious mind tells it how far to move like how many radii to move or more like percentage of radii and then it can expand again it knows how to expand the unconscious mind and it'll bump into something else maybe the same side maybe another side and so through this iterative process at a certain point it will have less and less room to move it'll never know what side it's

bumping into because it doesn't have that information but it'll just know that it's starting to bump in very often on you know which would presume that it's being closed in and so then the idea is that once you get within whatever accuracy you want because it never touches all three sides at once it only touches one side at a time um realistically but um when you get within 1% or whatever you want then you say okay I'm happy I'll stop and so then you can ask for example well what can you deduce about the shape of the triangle based on the interactions that you had blind like this you can also have like a learning process you know then you can maybe compare that with the original triangle uh you can um maybe also have a um a learning process where the conscious part that decides um how far to go like how how far to move this um it can um different policies can yield different results so it can maybe try to learn that so I don't know if that was like a my initial attempt to try to come up with a way of thinking about this language and seeing like how active inference could kind of help to apply it yeah at least I think that's interesting what you were saying and it there's some stuff I was reading by Michael Levan around like an axis of persu with different nested levels of intelligence that's kind of interesting to me that I think touches on what you're you're saying about different you

know I guess lower levels of um the system learning something and sending a signal out globally when there's some sort of an error um but I've been thinking about that those levels of persuasion because I think it speaks to from a communication perspective the way that we communicate we do it through unconscious associations we do it more rationally we do it emotively from a cultural perspective um but it like you know I think when it's most effective it's across all of those levels or that axis of persuasion that he's kind of talking about so is there such a thing as one of these models that's so simple it could be coded from a blank sheet of paper in I guess five minutes yes that's what the textbook level focuses on the textbook version is just five matrices with it in chapter 7 that's what this is what you build up to it's just starting from um Dynamic perception models partially observable models observations coming in this could be a sing binary variable this could be a three-state variable this could be continuous V like that's the work of modeling but then you have a hidden or latent State temperature in the room temperature on the thermometer mapping Matrix initial prior to kick the chain off transition Matrix then that's that's equivalent to a Colman filter like modeling then you add in some element of policy

selection is it just a binary uh with
two fixed outcomes or do you want to
model this or
that uh conditionality or
learning or or or uh
affordance State St space changing but
you know those are sort of those are
those are structures of models families
of models that you may on your on your
Intuition or or uh fun or whatever seek
to just work within a given model family
just because or not
realize um that you're with within a
given constraint um but it's just what
is it is it just being analytically just
here's like the space of all possible
policies of the organism and I'm about
to make a philosophical point or we
chose to just model the blood sugar not
engaging with other
biomarkers but this is these are the
matrices and the variables that you get
to a b c
d well I'd sort of like to see this
implemented in code uh is that available
yeah great question that's um again
that's what
chapter
6 gets to and I'll just
link I I know it's in the textbook group
um somewhere
hold this is a table of of
implementations if people want to add
and curate more it would be awesome
there's always
uh a lot we can add with these repos
um and if you again these are for people
who want to contribute open

source

resources Andor review them and annotate them with um like what features they have like which ones have planning which ones use this or that uh which ones have this C I just want like a a really really simple one you can figure out what it's doing by

inspection

yeah isn't doesn't appendix C mention like the T thing that you were talking about the t- maze is there is there code for that um there is mat lab code for that it's a uh in appendix C but we haven't in a prior textbook group ported it

but if people would like

to uh work on

that in in the times between

um meetings and uh get it working or just share what they where they got to that could be

awesome but making PDP

models and using expected free energy and variational free

um in the implementations and in this subsidi um

repo in the cognitive

repo there are there are it's just pdps with vfe and

EF also I want this is the slide from

the um the the three

PS a distinct rule for the distribution

P this is the same underlying

prior

distribution so if we were modeling like we're move we're interested in the movement data of thermophilic bacteria

so we're modeling the distribution of
their location with respect to some
fixed wall so we're going to have some
distribution for their
location it is in the first um frame the
probability of they're existing at that
location at that measured time it's a
it's a latent distribution smooth
analytically um approximable Etc
variational distribution implementable
with software tools for their location
it is also from the co cognitive
behavioral bacterial side as their
preference like our um expected
pragmatic
homeostatic firstness and tendency to um
of certain variables to return to a TR
States and it's the
Fitness in that it's the uh
ecological sampling
availability constituting differential
persistence phenotypically or
genotypically
but that just just that's one it's like
we're doing a Pon distribution from the
wall
any closing uh comments
if if people want to think about what
will make this early time slot in its
alternating you know next week at this
earlier time slot we will go to chapter
2 so if people are um want to just stay
at the same time continuing in this way
we've done it or
just do people have certain amounts of
time they want to set aside or certain
Milestones they want to reach then uh I
I mean I want it to be useful we can

make generative guides and

material

absurdly with people's feedback and help

if they want to make certain kinds of

models so it be it would be awesome to

work on that um but if people have some

uh preference for for what would be

good given that we all are here uh for

this time let's figure out so just like

email uh or or at the next meeting add a

note or we'll discuss it oh I just like

to say I think uh some more like coding

Workshop type elements would be good

yeah I I I totally

agree I'll be trying to code what I'm

doing so and I'm interested to talk with

others about you know how they code and

yeah it's like the low road is like the

show me the code Road high road is like

some is

um symbolic and analytical and axiomatic

uh then sort of qualitative

philosophical stances are included in

that sort of

combination and

uh I think we can we can have some cool

code models please just surface

whichever uh

desada you

see

and let's uh do it so I wanted to thank

uh for the time I apologize for my

connection and just to say like what I

learned was that like what am I trying

to predict I'm predicting where I think

the center of the triangle is and so

when I move the circle from The Edge I

have to make a prediction how far to

move it and that prediction could be wrong you know and I get feedback from the from the from the perimeter of the circle so it's a conversation between the predictions for the center of the circle and then the feedback that you get from the boundary of the circle that's getting closer to I think so and I guess for active inference you really need to understand like prediction error like what is the prediction what's the source of the error just random point about this circle triangle thing wouldn't you want to after it hits the wall wouldn't you want to grow it in a particular direction rather than move it in a particular direction well I want to have that dialogue so um so um the the two steps are you can move you can grow so the unconscious mind only knows how to execute this it doesn't and it knows when it hits something but it doesn't know anything really so this blind conscious mind has to has to have this uh communication that's what I want to model might it be better to have grow from the center and grow in a particular direction when well the point is that you don't know where the center is you're trying to find the center you're randomly inserted wait I thought you knew the center of the circle just not the triangle oh the center of the circle you know right so and you can only expand you can grow larger but you can

also grow a circle in a single Direction
rather than from the center of the
circle and and I think that would be a
faster implementation of what you're
yeah I'm not trying to do it faster I'm
trying to do slower so stupider like
what's the stupidest implementation
that's what I'm trying to
do most stupid in revealing thanks
though okay thank you bye